

Nexus HMS: Complete User Documentation

Version 1.0 | Enterprise Hospital Management System

This comprehensive guide provides end-to-end documentation for the Nexus HMS (Hospital Management System), covering all user roles, workflows, and system capabilities.

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System Overview

Nexus HMS is a comprehensive, FHIR-native, multi-tenant hospital management system designed for modern healthcare facilities. It integrates electronic health records (EHR), practice management, patient engagement, and revenue cycle management into a unified platform.

Key Features

- Multi-tenant Architecture:** Support for multiple hospitals/facilities
- Role-based Access Control (RBAC):** Granular permissions for different user types
- Real-time Dashboards:** Role-specific views with actionable insights
- Unified Patient Workspace:** 3-panel interface for comprehensive patient care
- HIPAA Compliant:** Built-in audit logging and data encryption
- Mobile Responsive:** Works on tablets and mobile devices
- API-First Design:** RESTful APIs for integrations

System Architecture

- Frontend:** React.js with Vite, Tailwind CSS
- Backend:** NestJS (Node.js) with TypeScript
- Database:** SQLite (development) / PostgreSQL (production)
- Cache:** Redis (optional, with mock support)
- Authentication:** JWT with HttpOnly cookies

Getting Started

Prerequisites

- Node.js 18+
- npm or yarn
- Docker (for database)
- Modern web browser

Installation and Setup

1. Clone the Repository

```
git clone <repository-url>
cd next-gen-hms
```

2. Start the Database

```
cd services/clinical-service
docker compose up -d
```

3. Configure Environment

The system auto-configures for development. Key settings in `.env`:

- `NODE_ENV=development`
- `PORT=3001`
- `DB_TYPE=sqlite`
- `USE_REDIS MOCK=true`

4. Start Backend Services

```
cd services/clinical-service
npm install
npm run start:dev
```

5. Start Frontend

```
cd frontend
npm install
npm run dev
```

6. Seed Demo Data

Visit: `http://localhost:3001/api/v1/auth/seed`

Accessing the Application

- URL:** `http://localhost:5173`
- Demo Login:** Use "Use Demo Account" button
- Manual Login:** Use seeded credentials (see User Roles section)

User Roles and Permissions

Nexus HMS supports multiple user roles with specific capabilities and access levels.

1. System Administrator (`hospital_admin`)

Capabilities:

- Full system access and configuration
- User management (create, edit, deactivate users)
- System settings and customization
- Audit log access
- Multi-tenant management

Demo Credentials:

- Email: admin@hms.local
- Password: admin123

Key Workflows:

- User onboarding and role assignment
- System configuration and maintenance
- Security policy management
- Compliance reporting

2. Physician/Doctor (doctor)

Capabilities:

- Patient consultation and charting
- Order entry (labs, imaging, medications)
- Prescription writing
- Clinical decision support access
- Patient history review
- Discharge planning

Demo Credentials:

- Email: doctor@hms.local
- Password: doctor123

Key Workflows:

- Daily patient rounds
- New patient consultations
- Medication reconciliation
- Diagnostic ordering
- Care plan documentation

3. Nurse (nurse)

Capabilities:

- Vital signs recording
- Medication administration (BCMA)
- Patient monitoring
- Care coordination
- Patient education
- Discharge instructions

Demo Credentials:

- Email: nurse@hms.local
- Password: nurse123

Key Workflows:

- Shift handoffs
- Medication administration
- Patient assessment
- Care planning
- Patient monitoring

4. Receptionist/Front Desk (receptionist)

Capabilities:

- Patient registration and check-in
- Appointment scheduling
- Queue management
- Patient information updates
- Basic billing inquiries

Demo Credentials:

- Email: receptionist@hms.local
- Password: receptionist123

Key Workflows:

- Walk-in patient registration
- Appointment booking
- Check-in/check-out process
- Patient information updates

5. Billing/Finance Staff (billing)

Capabilities:

- Invoice generation and management
- Insurance claims processing
- Payment processing
- Financial reporting
- Account reconciliation

Demo Credentials:

- Email: billing@hms.local
- Password: billing123

Key Workflows:

- Daily billing operations
- Insurance claims submission
- Payment posting
- Financial reconciliation

6. Patient (patient)

Capabilities:

- View personal health records
- Schedule appointments
- Access test results
- Communicate with care team
- Update personal information

Demo Credentials:

- Email: patient@hms.local
- Password: patient123

Key Workflows:

- Appointment scheduling
- Health record access
- Test result review
- Care team communication

 Clinical Workflows

Patient Registration Process

1. Access Registration
 - Navigate to "Quick Register" in sidebar
 - Or use Command Palette (Ctrl+K) → "Register"
2. Enter Patient Information
 - Full Name (required)
 - Date of Birth (required)
 - Gender (required)
 - Mobile Number (required)
 - Email (optional)
 - Address (optional)
 - Emergency Contact (optional)
3. Department Selection
 - Choose appropriate department
 - System assigns priority based on department
4. Complete Registration
 - Click "Complete Registration"
 - System generates unique MRN (Medical Record Number)
 - Patient added to department queue

Physician Consultation Workflow

1. Access Patient Queue
 - Login as physician
 - View dashboard with patient queue
 - Click on next patient
2. Patient Workspace Overview
 - **Left Panel:** Patient demographics, allergies, medications, recent vitals
 - **Center Panel:** Clinical notes, assessment, plan
 - **Right Panel:** Orders, prescriptions, test results
3. Clinical Assessment
 - Review patient history
 - Document chief complaint
 - Perform physical examination
 - Record assessment and plan
4. Order Entry
 - Search for labs, imaging, or procedures
 - Select appropriate tests
 - Add clinical justification
 - Submit orders
5. Prescription Writing
 - Search medication database
 - Select drug, dose, frequency
 - Check for drug interactions
 - Print or send electronically
6. Documentation Completion
 - Sign clinical note
 - Update patient status
 - Schedule follow-up if needed

Nursing Care Workflow

1. Daily Shift Start
 - Review assigned patients
 - Check pending tasks
 - Review care plans
2. Patient Rounds
 - Assess patient condition
 - Record vital signs
 - Update care plan
 - Document interventions
3. Medication Administration

- Review medication orders
- Perform 5 Rights check
 - Right patient
 - Right medication
 - Right dose
 - Right route
 - Right time
- Document administration
- Monitor for adverse reactions

4. Patient Education

- Review discharge instructions
- Educate on medications
- Answer patient questions
- Document teaching

Diagnostic Workflow

1. Order Processing

- Receive orders from physicians
- Prepare patient for procedure
- Perform diagnostic test

2. Result Entry

- Enter test results
- Review for critical values
- Flag abnormal results

3. Result Verification

- Physician reviews results
- Discuss findings with patient
- Update treatment plan



Administrative Functions

User Management

1. Access User Management

- Login as System Administrator
- Navigate to "User Management"

2. Create New User

- Click "Add User"
- Enter user details
- Assign role and permissions
- Set temporary password

3. Edit User Permissions

- Select user from list
- Modify role assignments
- Update access permissions
- Save changes

System Configuration

1. Access Settings

- Navigate to "System Settings"
- Configure hospital information
- Set operational parameters

2. Department Setup

- Add/edit departments
- Configure workflows
- Set access permissions

Audit and Compliance

1. Access Audit Logs

- Navigate to "Audit Logs"
- Filter by date, user, action
- Export reports

2. Compliance Reporting

- Generate HIPAA compliance reports
- Review access logs
- Monitor system usage



Financial Operations

Billing Process

1. Invoice Generation

- System auto-generates invoices from clinical activities
- Review charges for accuracy
- Add additional services if needed

2. Payment Processing

- Accept payments (cash, card, insurance)
- Post payments to patient account
- Generate receipts

3. Insurance Claims

- Submit claims electronically
- Track claim status
- Handle denials and appeals

Revenue Cycle Management

1. Dashboard Overview

- View accounts receivable
- Monitor claim status
- Track payment trends

2. Collections Management

- Identify overdue accounts
 - Send payment reminders
 - Process write-offs
-

System Administration

Backup and Recovery

1. Database Backup

- Automated daily backups
- Manual backup option
- Offsite storage

2. System Recovery

- Restore from backup
- Data validation
- System testing

Performance Monitoring

1. System Health

- Monitor server performance
- Database query performance
- API response times

2. User Activity

- Track concurrent users
 - Monitor resource usage
 - Generate usage reports
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Advanced Features

Command Palette (Ctrl+K)

- Global search and navigation
- Quick action triggers
- Patient search and access

Real-time Notifications

- Critical lab results
- Patient status changes
- System alerts

Mobile Access

- Responsive design for tablets
- Mobile-optimized workflows
- Offline capability (limited)

Integration Capabilities

- HL7/FHIR APIs
 - Third-party system integration
 - Data export/import
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Troubleshooting

Common Issues

Login Problems

- Verify credentials are correct
- Check account status (not deactivated)
- Clear browser cache

Slow Performance

- Check internet connection
- Clear browser cache
- Contact IT support

Data Not Saving

- Check form validation
- Verify required fields
- Contact system administrator

Support Resources

- **User Manual:** This document
 - **Help Desk:** support@hms.local
 - **Training Videos:** Available in system
 - **API Documentation:** http://localhost:3001/api/docs
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API Reference

Authentication Endpoints

- POST /api/v1/auth/login - **User login**
- POST /api/v1/auth/logout - **User logout**
- GET /api/v1/auth/seed - **Seed demo data (dev only)**

Patient Management

- GET /api/v1/patients - **List patients**
- POST /api/v1/patients - **Create patient**
- GET /api/v1/patients/:id - **Get patient details**
- PUT /api/v1/patients/:id - **Update patient**

Clinical Operations

- GET /api/v1/encounters - **List encounters**
- POST /api/v1/encounters - **Create encounter**
- GET /api/v1/vitals - **Get vitals**
- POST /api/v1/vitals - **Record vitals**

Administrative

- GET /api/v1/users - **List users**
- POST /api/v1/users - **Create user**
- GET /api/v1/settings - **Get system settings**

For complete API documentation, visit: <http://localhost:3001/api/docs>

Contact Information

- **Technical Support:** tech-support@hms.local
- **Clinical Support:** clinical-support@hms.local
- **Billing Support:** billing-support@hms.local
- **Emergency Hotline:** 1-800-HMS-HELP

This documentation is maintained by the Nexus HMS development team. Last updated: May 7, 2026</content>
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